

# Measuring emergency medical service (EMS) accessibility with the effect of city dynamics in a 100-year pluvial flood scenario

Mengya Li<sup>a</sup>, Mei-Po Kwan<sup>b,c</sup>, Jie Chen<sup>d</sup>, Jun Wang<sup>a,\*</sup>, Jie Yin<sup>a</sup>, Dapeng Yu<sup>e</sup>

<sup>a</sup> Key Laboratory of Geographic Information Science (Ministry of Education), School of Geographic Sciences, East China Normal University, Shanghai 200241, China

<sup>b</sup> Department of Geography and Resource Management and Institute of Space and Earth Information Science, The Chinese University of Hong Kong, Shatin, Hong Kong, China

<sup>c</sup> Department of Human Geography and Spatial Planning, Faculty of Geosciences, Utrecht University, 3584 CB Utrecht, the Netherlands

<sup>d</sup> State Key Laboratory of Resources and Environmental Information System, Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences, Beijing 100101, China

<sup>e</sup> Centre for Hydrological and Ecosystem Science, Department of Geography, Loughborough University, Loughborough LE11 3TU, UK

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## ABSTRACT

Emergency medical service (EMS) is important for rescuing victims suffering from life-threatening illnesses or accidents, and is highly time-sensitive by nature. Many uncertain contexts in the urban environment can prolong EMS response time and deteriorate its performance. Using the enhanced two-step floating catchment area (E2SFCA) method, this study measures EMS accessibility with the effect of a regular uncertain context (i.e., the city dynamics like time-varying population and traffic) and an irregular uncertain context (i.e., an extreme pluvial flood event which can cause extensive road closures). The results indicate that, in the central urban area of Shanghai, mid-west areas with denser populations have higher accessibility than eastern peripheral areas. Flooding can cause a remarkable decline of accessibility which falls to the lowest point slightly earlier than the time when the worst road connectivity emerges. The night time exhibits better accessibility than especially the peak hours during the daytime. The GWR results reveal that increasing facility richness and road density while decreasing flood-induced road closures have a positive effect on EMS accessibility. The study indicates that both regular and irregular uncertain contextual factors can influence EMS accessibility in a highly complex manner. Carefully taking these uncertainties into account would enable EMS planning in other contexts and regions to face the enormous challenges posed by the changing climate and increasingly complex urban environment.

## 1. Introduction

As one of the pivotal public health services, emergency medical service (EMS) plays a vital role in reducing the rate of morbidity and mortality, especially in the cases of accidents, sudden illnesses and urgent situations (e.g., life-threatening trauma, cardiac arrest, strokes, and childbirth) (Aringhieri et al., 2017; Liu et al., 2019). However, many uncertain contexts in the urban environment can prolong EMS response time and deteriorate its performance. Overlooking the contextual effects may lead to problematic evaluation of EMS accessibility and misinformed policy interventions (Hawthorne & Kwan, 2012).

Accessibility is an important indicator for evaluating EMS response performance. As widely used in disciplines like transportation, urban

planning and geography, accessibility has a variety of definitions (Geurs & van Wee, 2004; Hansen, 1959; Hu & Downs, 2019; Kwan, 1998; Le Vine et al., 2013). Basically, how accessibility is measured depends on how it is conceptualized. According to Penchansky and Thomas (1981), a multi-faceted concept for health care access involves five dimensions (affordability, acceptability, availability, accommodation, and geographic accessibility), which respectively denote the cost, satisfaction, sufficiency, appropriateness, and travel impedance of (using) health care services. Among them, geographic accessibility (also known as spatial or physical accessibility) attracts considerable attention from health- and transport geography studies, where it is given a broader meaning referring to the availability of and proximity to health care services (Luo & Wang, 2003).

\* Corresponding author at: Key Laboratory of Geographic Information Science (Ministry of Education), School of Geographic Sciences, East China Normal University, 500 Dongchuan Road, Minhang District, Shanghai 200241, China.

E-mail addresses: [wooddream2009@163.com](mailto:wooddream2009@163.com) (M. Li), [mpkwan@cuhk.edu.hk](mailto:mpkwan@cuhk.edu.hk) (M.-P. Kwan), [chenj@reis.ac.cn](mailto:chenj@reis.ac.cn) (J. Chen), [jwang@geo.ecnu.edu.cn](mailto:jwang@geo.ecnu.edu.cn) (J. Wang), [jyin@geo.ecnu.edu.cn](mailto:jyin@geo.ecnu.edu.cn) (J. Yin), [d.yu2@lboro.ac.uk](mailto:d.yu2@lboro.ac.uk) (D. Yu).

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In view of this two-dimensional definition, the two-step floating catchment area (2SFCA) method has gained great popularity in relevant studies because it measures availability and proximity simultaneously by considering demand (e.g., population size), supply (e.g., number of EMS facilities), and the spatial interaction between them (Wang, 2017). As the name implies, 2SFCA evaluates accessibility through two steps. The first is assigning a supply-to-demand ratio to each supply node, and the second is calculating the accessibility of each demand node. The catchment area is a constraint of time or space that determines whether a pair of supply and demand nodes can spatially interact. Currently, there are several types of improvements to the original 2SFCA method (Wang, 2012; Tao and Tao & Cheng, 2016), mainly focusing on the distance decay function (e.g., enhanced 2SFCA) (Luo & Qi, 2009), the delimitation of catchment areas (e.g., dynamic 2SFCA) (McGrail & Humphreys, 2014), the competition among demand nodes or supply nodes (e.g., modified 2SFCA) (Delamater, 2013), the travel means of the demand side (e.g., multi-mode 2SFCA) (Mao & Nekorchuk, 2013), and actual human travel behavior (e.g., E2SFCA-ICA) (Pan et al., 2018).

With continuous improvements, the 2SFCA method has been shown to be capable of yielding increasingly reliable and accurate measurement of health care accessibility; but in the context of EMS, new difficulties arise. Unlike non-emergency problems, EMS response is highly time-sensitive. Higher response time may seriously reduce the survivability of victims to be rescued (Amorim et al., 2019). Measuring EMS accessibility, in particular, should incorporate the uncertainties that may cause fluctuations in response times. However, most empirical studies to date conceptualize accessibility in static terms and ignore the variations in demand, supply, and travel impedance over time and space, and thus suffer from the uncertain geographic context problem (UGCoP) (Kwan, 2012, 2018). Solutions for this problem should be based on a good understanding of what the uncertainties are or where they stem from. The primary purpose of this study is to answer the following question: How do the typical uncertain contextual factors in urban areas (e.g., the time-varying population and traffic, an extreme pluvial flood which can cause extensive road closures), influence EMS accessibility? Carefully taking these uncertainties into account can help improve the measurement of accessibility in complex spatial and temporal contexts and provide more credible policy and practical recommendations for emergency management to meet greater challenges against a backdrop of the changing climate with more uncertainties.

## 2. Literature review

Herein, we reviewed two types of uncertain contexts, which have been discussed in a limited number of studies, encountered by the measurement of EMS accessibility.

The first type of uncertain context is *regular* uncertain context, namely the daily changes of traffic and demand due to millions of individuals' mobility, which can simply be referred to as city dynamics. Variations in EMS demand have been noted very early (Cusimano et al., 2010; Fransen et al., 2015; Jamtsho et al., 2015), but the lack of efficient tools to capture mobility patterns makes traditional studies rely heavily on census data to represent demand statically. Xia et al. (2019) claimed that using census data may overestimate EMS demand in the rural areas and underestimate it in downtown during the daytime, and they proposed a spatiotemporally enhanced 2SFCA (ST-E2SFCA) method to measure EMS accessibility drawing on GPS dataset to depict the hourly change of demands. Andersson et al. (2020) utilized a location-allocation model called maximum expected performance location problem with heterogeneous regions (MEPLP-HR) to reveal how EMS performance varies during the day/week by taking time-varying demand into account. Additionally, some paid attention to the rising demand for emergency services induced by long-term population growth over the past decades or in a future scenario (Kiran et al., 2020; Yang & Mao, 2018). Aside from demand, large volumes of people flow also cause dynamic traffic conditions, whereas the effect of traffic status on

EMS accessibility was less studied. Chen et al. (2016) used GPS trajectories data to calculate the vehicle speed during normal and peak hours in a day and found that traffic congestion might have a large influence on EMS accessibility. Hu et al. (2020) applied a traffic simulation model to estimate the EMS response time under free-flow and congested road conditions and similarly concluded that the spatial accessibility would be reduced when traffic congestion is considered. More rarely are studies considering a combination of varying demand and travel time. An exception is Amorim et al. (2019) which used official EMS data (over 30,000 records containing information on the type of emergency, timestamp, and address) to study the impacts of daily changes in traffic and demand on EMS strategic decisions; but both the data and the method have limited transferability.

The second type of uncertain context is *irregular* uncertain context, which refers to traffic accidents, natural disasters, and other unexpected events that tend to cause sudden traffic disruptions or congestions. Here we highlight the impacts of extreme flooding, which is among the top-ranked threats to human habitats, especially in urban areas with dense populations and infrastructures (Yu et al., 2020). Extreme flood events on one hand may lead to a sudden rise in demand for emergency services, as people are exposed to greater danger brought by flooding and its potential cascading consequences. On the other hand, roads rendered impassable by flood and the concomitant traffic congestion may cause failure to reach demand locations within the mandatory response times (Coles et al., 2017; Yin et al., 2018). Several studies have paid attention to the impacts of severe flooding on EMS. For example, Coles et al. (2017), Yu et al. (2020) and Yin et al. (2018), Yang et al. (2020) mapped the spatial coverage of EMS stations within the legislated time frames or calculated response times from EMS stations to high-priority demand nodes (e.g., vulnerable groups) under different flood scenarios in the UK and Shanghai respectively. By considering pluvial flood probability, Rizeei et al. (2019) assessed the coverage of emergency response centers (including EMS) and projected optimal service locations for vulnerable inhabitants with redundant or late emergency response. But these studies still have some shortcomings. Whereas proximity is measured using a simple coverage definition (e.g., services within 7-, 10- and 15-minute response times), availability (i.e., the competition for services) is often overlooked. Besides, when calculating response times, ambulances' speeds are set through sensitivity test (e.g., 20%, 50% or 80% of the speed limits) or adding a fixed deviation (e.g., 12 mph below/above the speed limits) to address the traffic conditions, but in fact congestion also varies spatially.

Despite that each of the aforementioned uncertainties has been discussed in previous studies, to the best of our knowledge, no research has explored and addressed a combination of them in the measurement of EMS accessibility. Thus, the major contribution of this study is to comprehensively discuss the UGCoP in the evaluation of EMS accessibility. By taking advantage of spatiotemporal big data (e.g., the mobile signaling data and real-time traffic data) as well as high-resolution flood simulation results, we can measure EMS accessibility while taking into account the effects of time-varying demand (population) and travel impedance (speed and road connectivity) at a very fine scale. Besides, we further explore whether the effects of the contextual factors are statistically significant in order to provide more reliable and effective suggestions for improving EMS management in urban areas. In addition, past studies have found that ignoring contextual uncertainties like variations in population distribution and traffic can lead to serious errors in accessibility evaluations (e.g., Ren et al., 2014). This means that under certain emergency situations (e.g., earthquakes), some areas may have unexpectedly low accessibility that may put a large number of needy people at risk. Thus, the method used in this study highlights how some of these uncertainties can be taken into account and why doing so is essential for emergency services in other regions of the world. This emphasis thus also contributes to the emerging international literature on the effects of uncertainties encountered when examining a wide range of issues involving human-environment interactions besides

healthcare access (such as environmental health; e.g., Kwan et al., 2019; Helbich et al., 2021).

The remainder of the paper is structured as follows. The next section describes the workflow of the study (including the case study site, data source, method of accessibility measurement). Sections 4 and 5 present the result and discussion respectively. Section 6 concludes with a summary of the key findings, along with some policy recommendations and reflects on the research limitations.

### 3. Data and methods

A brief workflow is illustrated in Fig. 1. Slightly different from the chronological order of the workflow, the following subsections first give an introduction to the study area and the EMS system in it, and then successively describe the detailed methods of (1) measuring accessibility using E2SFCA, (2) depicting the daily changes of the population and traffic (the regular context), and (3) defining flood-induced road closures (the irregular context).

#### 3.1. Study area

Shanghai is the core of the Yangtze River Delta Megalopolis, which is an increasingly important region for global cooperation in technology, finance, and trade. It is also the top contributor to the Yangtze River Delta Alliance of Pre-hospital Emergency formed under the strategic framework of regional integration (Shanghai Municipal Health Commission, 2020).<sup>1</sup> Whereas being a world-class megacity with the ability to provide superior health care service, Shanghai still has room for improvements in EMS. Before the 11th Five-year Plan (2006–2010), there is only 58 EMS stations and 240 ambulances in the whole city (Shanghai Medical Emergency Center).<sup>2</sup> Now, these numbers have increased to more than 160 and 900 respectively (Youth Daily, 2018-11-20).<sup>3</sup> According to the Shanghai Municipal Health Commission (2020), every two subdistricts (comprised of several neighborhoods) share one EMS station, and every 30,000 people share one ambulance. In 2018, the ambulances dispatched by the Shanghai EMS totaled about 0.8144 million and altogether rescued 0.7397 million people.<sup>4</sup> Although the Commission has claimed that by the year 2020 the average service radius and response time should be lower than 3.5 km and 12 min respectively, this is a challenging goal for areas with sparse EMS stations or bad traffic.

The left part of Fig. 2 shows the population density based on the 2010 Census (which is static and not used for measuring accessibility). This study only focuses on the central urban area (CUA) where 69 EMS stations (over 40% of the entire city) are located. The CUA is a geographic area separated from the suburbs by the Outer Ring road, accounting for about 10% of the city's territory but supporting almost half of its population and economic activities. Millions of residents' daily activities for various purposes produce large volumes of people flow and inevitably busy traffic. The difference between the daytime and nighttime population in the CUA may reach 0.5 million during a single day.

Besides, Shanghai is a typical coastal city with low-lying topography and a subtropical monsoon climate and is often hit by rainstorms along with varying degrees of pluvial flooding. During the process of rapid urbanization, Shanghai has experienced an extensive transformation of natural land cover to built-up areas with impervious surfaces and inadequate drainage systems (Huang et al., 2017; Yin et al., 2015). The highly urbanized CUA in particular is quite prone to pluvial flooding if the ongoing climate change brings heavier rainfall. All these facts

indicate that it is necessary and appropriate to investigate the effects of regular city dynamics and irregular flood-induced interruptions on EMS in the CUA of Shanghai, to provide more reliable practical recommendations for the municipal EMS system as well as experiences for other cities in the world.

#### 3.2. Measuring EMS accessibility

The E2SFCA method is adopted in this study to measure EMS accessibility, and the distance decay behavior is described by a continuous function. Like the classic 2SFCA, the E2SFCA also calculates accessibility through two steps. The first is assigning an initial ratio of supply to demand to each supply node. For each EMS station  $j$ , search all demand locations that are accessible within a threshold time  $t_0$ . Then the supply to demand ratio  $R_j^T$  at time  $T$  can be computed as:

$$R_j^T = S_j / \sum_{i=1}^n D_i^T f_{t(ij)} \quad (1)$$

where  $S_j$  is the supply value at station  $j$ ;  $D_i^T$  is the demand value at location  $i$  ( $i = 1, 2, \dots, n$ ). In the study area, there are 69 EMS stations and 718 demand locations. Each demand location represents the centroid of a 1 km  $\times$  1 km grid, in which the time-varying population is estimated.  $t(ij)$  is the access time between  $i$  and  $j$  ( $t(ij) \leq t_0$ ), and  $f_{t(ij)}$  is the distance decay function. The second step is to search all EMS stations that are accessible to demand location  $i$  within the same threshold time  $t_0$ , and sum up the initial ratios of supply to demand ( $R_j^T$ ,  $j = 1, 2, \dots, m$ ) as the accessibility of location  $i$  at time  $T$ :

$$A_i^T = \sum_{j=1}^m R_j^T f_{t(ij)} \quad (2)$$

A larger value of  $A_i^T$  indicates a higher level of accessibility. Here, four temporal scenarios are considered:  $T = 2:00, 9:00, 14:00$ , and  $18:00$  (24-h time) respectively represent the nighttime, the morning peak, the daytime leisure hours, and the evening peak. Note that  $f_{t(ij)}$  takes the form of a modified Gaussian function ( $e^{-t(ij)2/\alpha}$ ). The parameter  $\alpha$  determines its rate of decline and the point where it approaches zero (Kwan, 1998). In light of Shanghai CUA's practicality, suppose the maximum mandatory EMS response time is 15 min ( $t_0$ ), then  $f_{t(ij)}$  approaches zero when  $t(ij)$  approaches 15 min, based on which  $\alpha$  is computed.

Once the measuring method is set up, we are ready to define the input data: the supply of station  $j$  ( $S_j$ ), the demand of location  $i$  ( $D_i^T$ ), and the travel impedance  $t(ij)$  between them. The population is a good indicator of demand if the disparity of morbidity is ignored. However, due to a lack of detailed information about the EMS providers (e.g., the standing professionals and ambulances), we cannot estimate how supply may vary from station to station. Thus, all  $S_j$  are assumed to be 1 (unit). Travel impedance is mainly determined by vehicle speed and road connectivity, where travel speed depends on the real-time traffic and road connectivity is affected by the number and distribution of road closures as a result of the 100-year pluvial flood. Procedures for estimating the time-varying demand (i.e., population) and travel impedance (i.e., traffic and road connectivity) are described as follows.

#### 3.3. Definition of city dynamics

##### (1) Population

To capture the mobility patterns of the residents in the study area at fine time scales, we use a mobile communication signaling dataset provided by the China Mobile Company to estimate the hourly changes in population at a spatial resolution of 1 km. The original dataset covers 17 million mobile phone users for one day in 2012. Each record includes information about the user ID, timestamp, and X- and Y-coordinate. See Li et al. (2017) for more data cleaning and processing details. According

<sup>1</sup> <http://wsjkw.sh.gov.cn/xwfb/20191025/0012-66065.html>.

<sup>2</sup> <http://www.sh120.sh.cn/jjzx/n4/n15/u1ai401.html>.

<sup>3</sup> [http://www.why.com.cn/epaper/webpc/qnb/html/2018-11/20/content\\_74997.html](http://www.why.com.cn/epaper/webpc/qnb/html/2018-11/20/content_74997.html).

<sup>4</sup> <http://wsjkw.sh.gov.cn/2018wsjzkzkg/20190527/0012-64097.html>.



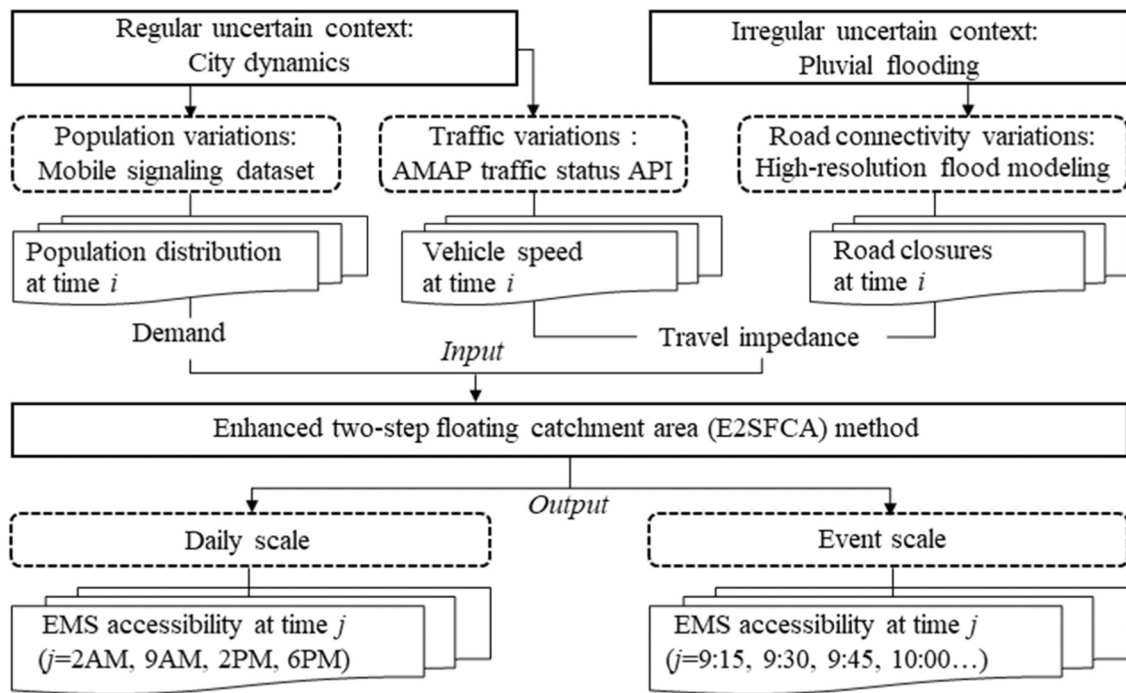


Fig. 1. Workflow of this study.

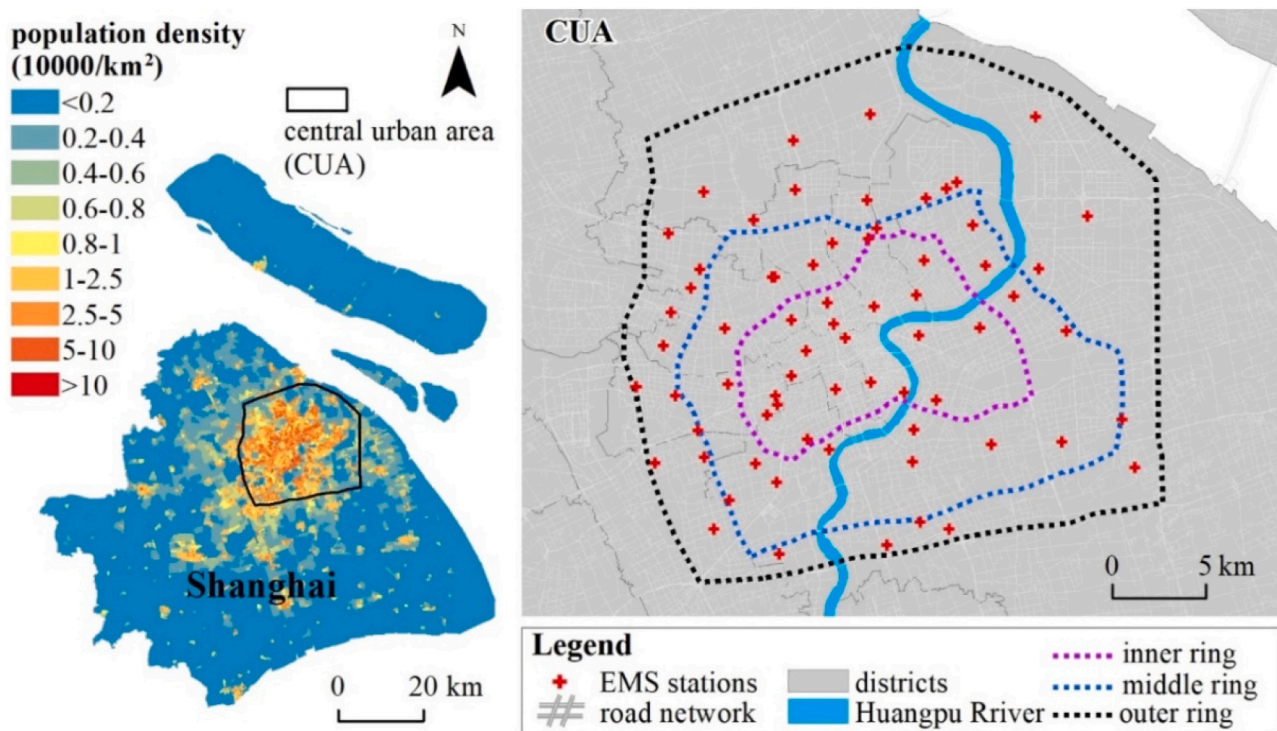


Fig. 2. Study area and the locations of concerned EMS stations.

to the official market penetration (65%) of the company, the population is estimated based on the number of China Mobile subscribers. The population size in each 1 km grid represents the demand of it.

The results show that the total population inside the CUA at 2:00, 9:00, 14:00, and 18:00 is about 11.12, 11.46, 11.56, and 11.59 million respectively. Fig. 3a displays an example of the population distribution at 9:00. People are denser inside the Inner Ring than in the peripheral areas. Two sections (highlighted in cyan) in the direction of west-to-east

(WE) and south-to-north (SN) are selected to reveal the temporal variations. From the graph in Fig. 3b we can see that the spatial distribution follows the same pattern in the four temporal scenarios (respectively displayed by solid- and dotted lines). People mainly gather at the center that corresponds to the middle part of the X-axis, and the peaks at 2:00 and 9:00 (the orange and blue lines) are evidently lower than those at 14:00 and 18:00 (the pink and green lines). However, the total population at 9:00 is much closer to 14:00 and 18:00. Generally, the highest

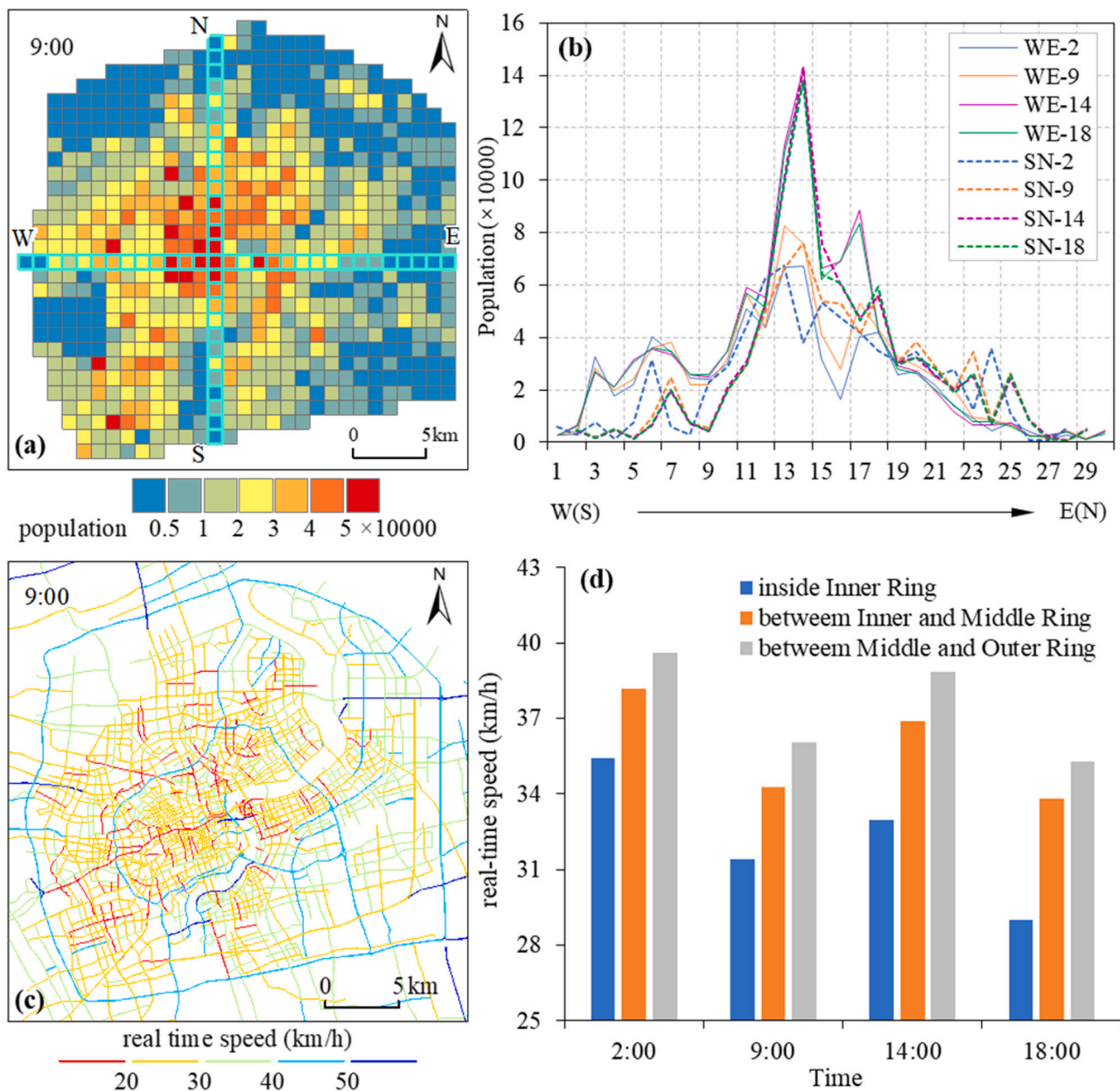


Fig. 3. The daily changes of population distribution (a, b) and real-time traffic (c, d).

demand for EMS occurs during the evening peak hours, and the spatial disparity in demand is more remarkable later in the day when crowds of people (e.g., commuters) have poured into the CUA, especially the area inside the Inner Ring.

## (2) Traffic

Real-time traffic is extracted from AMAP (<http://ditu.amap.com/>) through a web crawling Python program using the map's web service API. Specifically, five workdays' (May 18–22, 2020) traffic data at the time of 2:00, 9:00, 14:00, and 18:00 is collected and saved as shapefiles with 5 attribute fields: *road name*, *status* (describing the congestion degree), *direction* (from one to the other end of the road), *angle* (deviation from the horizontal, similar to the use of *direction*), and *speed* (the real-time vehicle speed on the road). Then, for each temporal scenario, the crawled data are merged to calculate a 5-day average speed for each road. Likewise, Fig. 3c displays an example of the spatially uneven traffic status at 9:00.

Next, the average speed layer is joined to a basic road dataset provided by the Key Laboratory of Geographic Information Science (Ministry of Education) in East China Normal University. The road dataset includes a total of 9029 road links and has more attributes like *road name*, *type* (elevated/expressways, primary roads, secondary roads, local roads), *length*, *direction* (one-way or two-way), *turn restriction* (global turns or restricted turns), and *legislated speed limits*. There are 5038–5095 matching records (a record means a road link) when joining the speed layer to the basic road layer, and the matched length accounts for 60%–65% of the entire road network. As for the unmatched records, most (82%) of them are local roads with a speed limit of 20 km/h. On these roads, the vehicle speed is set as 70% of the allowed maximum speed to address the traffic conditions. In other words, the real-time traffic on most high-level roads (e.g., the primary roads), which are more likely to be used by ambulances, can be captured. Finally, under normal scenarios, the average speed of the whole network at 2:00, 9:00, 14:00, and 18:00 is 33.17 km/h, 27.96 km/h, 31.56 km/h, and 26.06 km/h respectively. The bar chart in Fig. 3d shows that the real-time speeds



differ not only between time scenarios but also geographic locations. The speed in areas between the Outer Ring and the Middle Ring can be 10.6%–17.8% higher than that inside the Inner Ring. The time of 2:00 has the highest global speed and the smallest spatial disparity.

### 3.4. Producing flood scenarios

This part of the study is the same as that in Li, Kwan, et al. (2018), and thus only some critical model information is provided here. The external driving force of the flooding is a 1-in-100-year rainstorm. The Shanghai Rainstorm Intensity–Duration–Frequency (IDF) formula (*Standard of Rainstorm Intensity Formula and Design Rainstorm Distribution* (DB31/T-1043), 2017) is used to calculate the hourly rainfall, and the Chicago Design Storm method (Keifer & Chu, 1957) is used to generate the rainfall hyetograph (see the black dotted line in the graph in Fig. 4a). Another important input is the topography data that determines the surface flow direction and the inundation depth. A Digital Surface Model (DSM) dataset with a resolution of 5 m (which is about the width of the narrowest single-lane road and ensures that flooding details over the road network can be detected) is used. Flood modeling is implemented

using FloodMap-HydroInundation2D (Yu & Coulthard, 2015). Previous simulations have been performed with the same model to evaluate the influence of land subsidence on pluvial flooding (Yin et al., 2016), and it has been calibrated for the same site of this study and demonstrated to have good predictive accuracy. Therefore, we employ a group of model parameters according to Yin et al. (2016) to generate the target flood scenario. The model is run for 4 h until the surface flow reaches a steady state, whereas the rain only falls in the first hour.

#### (1) Criterion for road closure

There has been research on vehicle stability in floodwater showing that, in fast-flowing water with a velocity of 3 m/s or greater, it takes only 15 cm of floodwater for a small vehicle to become unstable, and 30 cm for four-wheel-drive vehicles (Molenaar et al., 2014; Smith et al., 2018). Ahmed et al. (2020) even reported that the shallowest water depth responsible for a fatal incident (drowning or electrocution) can be only 20 cm. Therefore, in some countries (e.g., the US, the UK, and Australia), 25–30 cm is adopted as a criterion for road closure as prescribed in traffic rules. If being called in an extreme flood scenario, the

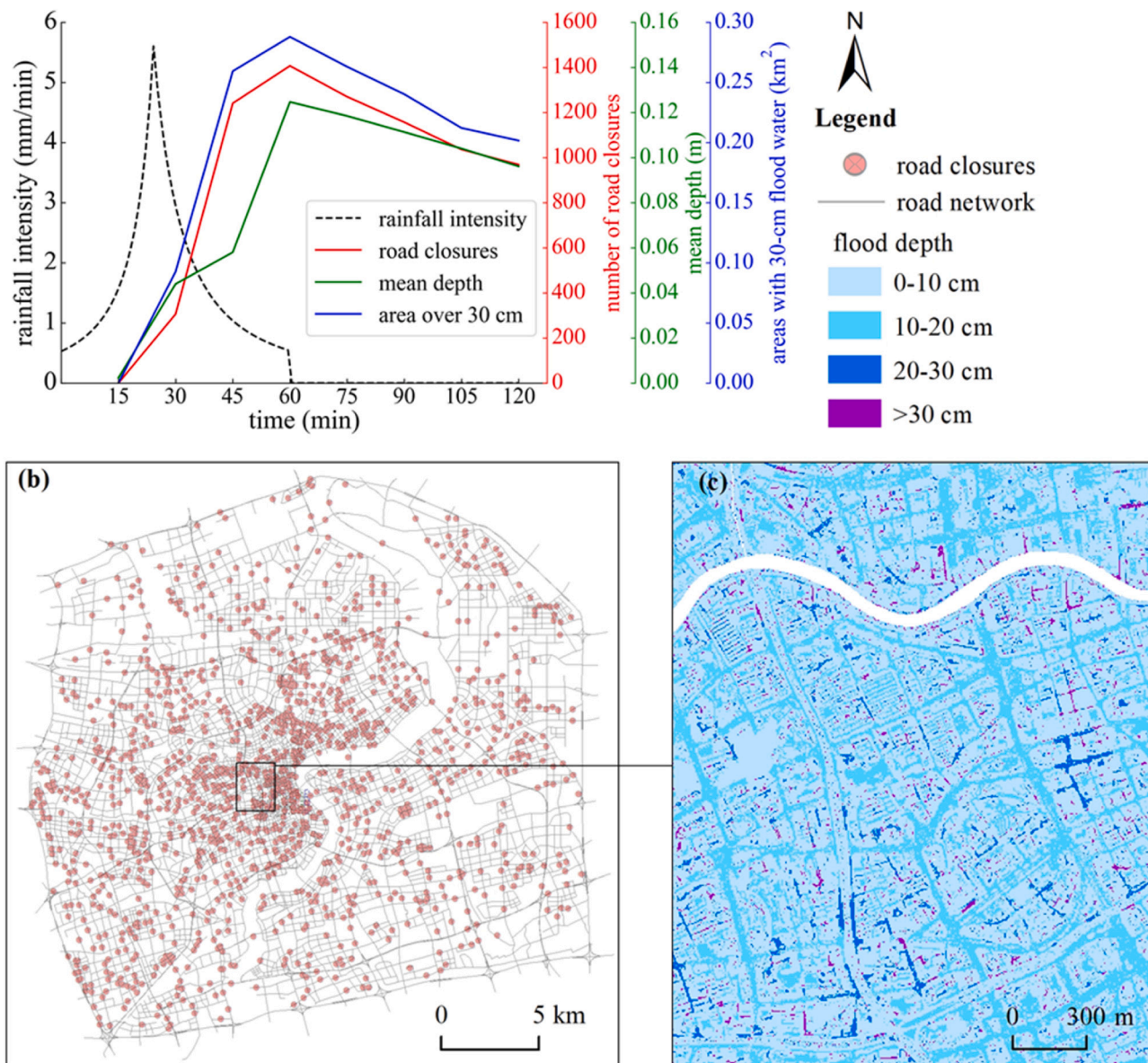


Fig. 4. The number of road closures and floodwater depth/extent at different flooding stages.

EMS providers no doubt need to take some risk when driving through the floodwater to reach the emergency sites. To balance their safety and duty, and ensure that an ambulance will not break down on the way to victims to be rescued, roads inundated by water over 30 cm are set as vehicle-impassable roads, which should be strictly avoided when routing ambulances. Note that the criterion of 30 cm is also used in some published works conducted in Shanghai that study the impact of floods on urban traffic or other transport-related issues (Li, Huang, et al., 2018; Li, Kwan, et al., 2018).

In view of the maximum EMS response time ( $t_0$ ), the modeling results are output every 15 min to reveal the spatio-temporal variations of water extents and depths during the entire simulation. There are extensive inundations of 10–20 cm on the roads. The distribution of inundations of 20–30 cm also routes along the roadside curbs but exhibits less continuity (see the zoomed-in map in Fig. 4c). As shown in Fig. 4a, the mean depth, areas with over 30-cm flood water, and the number of road closures all reach their peaks during the fourth stage (45–60 min), showing a relatively steep rise before the peak and a steady and slow drop after it. Fig. 4b displays the locations of road closures during the peak stage. Except for the beginning 0–15 min, the number of road closures in the subsequent 7 stages are 306, 1241, 1407, 1270, 1157, 1036 and 969, respectively.

## (2) The impacts of flooding on traffic speed

Travel impedance depends on both real-time traffic status and road connectivity. While pluvial flooding produces road closures in the road network, it may also change vehicle speed due to bad visibility caused by the rainstorm or spontaneous slow-down on the wet and slippery road. If the optimal path connecting an EMS station ( $j$ ) and a demand location ( $i$ ) consists of  $N$  road links, and it takes  $t(ij)$  for the ambulance to reach  $i$  after departure,  $t(ij)$  can simply be computed in the form of

$$t(ij) = \sum_k^N l_k / (b \times c \times v_k) \quad (3)$$

where  $l_k$  is the length of road link  $k$ , and  $v_k$  presents the speed on it.  $v_k$  equals the real-time speed if link  $k$  has a matching record in the speed layer crawled from AMAP; otherwise equals 70% of the speed limit. Besides, considering the specialty of ambulances that has a partial exemption from the ordinary traffic rules (e.g., move on the wrong side of the road or run through a red light) and that other vehicles may move over for an ambulance, parameter  $b = 1.25$  is employed to model that ambulances can go faster than the average speed on link  $k$ . Lastly,  $c$  is used to model the influence of varying rainfall intensity at different flooding stages. For the first and the fourth time intervals (0–15 min and 45–60 min), which are the beginning and ending stages with lower rainfall intensity,  $c$  equals 0.7. For the second and the third intervals (15–45 min), which are the peak stages with higher rainfall intensity,  $c$  equals 0.5. From the fifth to the eighth interval (60–120 min), when the rain has stopped whereas most roads are still covered by floodwater formed in the first hour,  $c$  equals 0.9.

## 4. Results

Altogether, we evaluated the EMS accessibility under 36 scenarios based on different combinations of the 4 temporal scenarios (2:00, 9:00, 14:00, and 18:00) and 9 flood condition scenarios. The latter refers to 1 normal scenario (NS) without flooding and 8 flood scenarios corresponding to 8 stages of the flooding. Meanwhile, for each flood scenario, we calculated the mean of 8 accessibility values (i.e., FS-mean) as the final assessment to be compared with the normal scenario. Since the research is designed based on the important idea of the UGCop, the purpose of utilizing different scenarios is to present the changes of uncertain geographic contexts over time and space.

### 4.1. EMS accessibility variations at the daily scale

Fig. 5a displays the spatial distribution of EMS accessibility under the normal and flood scenarios at the time of 2:00, 9:00, 14:00, and 18:00. The four maps in the first row (NS-2, 9, 14, 18) share similar spatial patterns, which suggests that, under non-flood conditions, the central areas have higher accessibility than the peripheral areas, and accessibility of the western part is better than the eastern part. From Section 3.3, we know that the population is also distributed inside the Inner Ring and the south-western quadrant, indicating greater demand in these areas. Interestingly, although the competition for service is one of the focal points considered in accessibility measurement, locations that are most populated have a higher but not lower level of accessibility compared to locations that are least populated. However, looking at the 69 EMS stations, 52 stations (over 75%) are located on the western side of Huangpu River, while only 17 are on the eastern side. Thus, the uneven distribution of service providers may be the primary contributing factor to the current spatial disparity of accessibility.

The four maps in the second row (FS-2, 9, 14, 18) of Fig. 5a reveal that EMS accessibility has experienced a remarkable decline in all flood scenarios. In particular, areas with accessibility values  $>15$  shrink considerably. Since network connectivity is reduced by 3.4%–15.6% due to road closures, an ambulance may have to drive along a new route instead of the best one which can save more time on the way to victims. Worse yet, some demand locations may become completely inaccessible within the maximum response time of 15 min. Further data analysis showed that areas with accessibility  $\geq 13$  (high accessibility) decreased by 77.0%–86.0%, whereas areas with accessibility  $\leq 3$  (low accessibility) increased by 48.6%–54.4%. A few locations exhibit higher accessibility in the flood scenarios. This somewhat counterintuitive phenomenon should also be explained by the competition for facilities, since there are locations gaining service from those that are losing.

From Fig. 5(a, b) we can see the evident difference between a normal and a corresponding flood scenario. But interestingly, the scatters in Fig. 5c demonstrate good correlations between them in all temporal scenarios. This result may be explained by the fact that the flooding has caused a universal decrease in accessibility but has not largely changed the spatial pattern of it. Most locations with higher accessibility in the normal scenario may also be more accessible in the flood scenario and vice versa.

### 4.2. EMS accessibility variations at the event scale

To reveal the accessibility variations at the flood event scale, we calculated the average accessibility over the entire study area during each 15-minute stage of the flooding. Besides, the numbers of accessible locations (NoALs) within 5 different time frames (5, 8, 10, 12, and 15 min) are counted. The horizontal axis in Fig. 6 represents the 8 stages, while the left blue vertical axis represents the accessibility value, and the right red vertical axis represents the NoALs.

The four charts in Fig. 6 suggest that changes in accessibility and the NoALs share similar trends in four temporal scenarios. Both indices fall gradually during the first three stages and to the lowest point at the end of the third stage (45 min). Then, they rise towards the original levels during the fourth and fifth stages (45–75 min) and increase very slightly in the last three stages (75–120 min). Within a response time as short as 5 min, only a limited number (less than 30%) of locations can be reached by EMS professionals, which neither shows notable variations during different stages of the flooding, especially in the daytime peak-hour scenarios. By contrast, locations that are accessible within 8, 10, 12, and 15 min present a similar degree of variability with the accessibility levels. Note that within the maximum response time (15 min) with non-flood conditions, over 90% of locations can be reached in the night time and daytime leisure hours, and approximately 87% in the daytime peak hours, which decreased to 70.1%–74.5% and 62.3%–64.1% respectively in the third flooding stage.



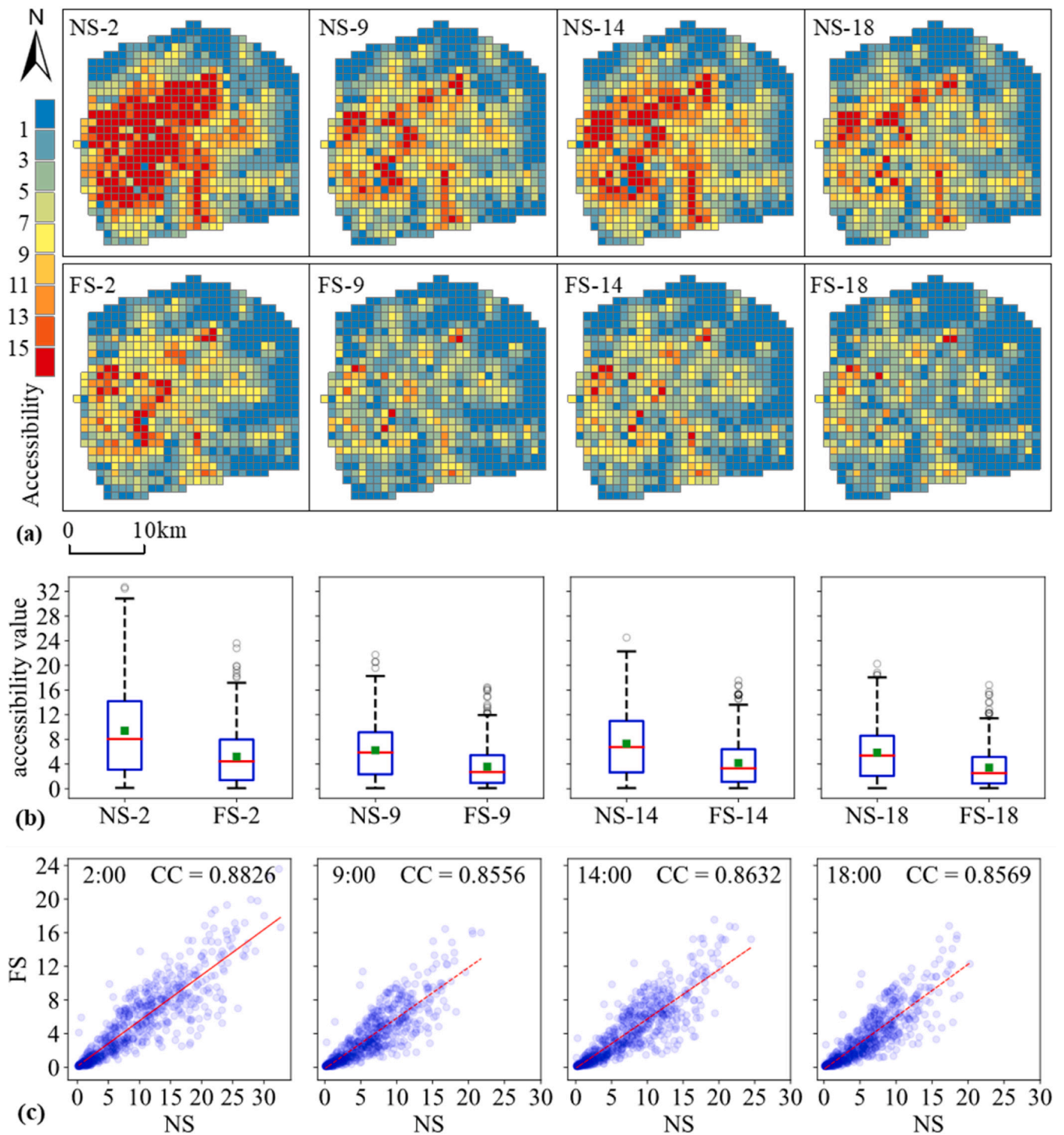


Fig. 5. Comparison of EMS accessibility in the normal (NS) and flood scenarios (FS).

(a: accessibility at 2:00, 9:00, 14:00, and 18:00 respectively, b: comparison of the 5 quintiles of accessibility values; c: correlation between accessibility under the normal and flood scenarios).

A similar outcome can be seen in Table 1. Here we set the average accessibility in the normal scenario at 2:00 (NS-2) as the baseline and then respectively calculate the root mean squared error (RSME) values between the other 35 scenarios and the baseline. As can be seen from the table, higher RSME values ( $>8$ ) emerge during 15–60 min, the three stages that undergo the largest reduction of accessibility. In the daytime peak scenarios (9:00 and 18:00), the maximum RSME (highlighted in bold) values appear at the second stage, while in the night time and daytime non-peak scenarios (2:00 and 14:00), they appear at the third

stage. Note that the second and the third stages are both half part of the rainfall peak. Additionally, the coefficient of variation (CV) value is employed to reveal the spatial variability of accessibility. CV values under the normal scenarios are less than 0.8 while CV values under most flood scenarios are higher than 0.8, indicating a greater EMS access variability with the effect of flood. The maximum CV values also show up at the third stage, followed by that in the second and fourth stages.

Taken together, Fig. 6 and Table 1 suggest that the third flooding stage has the lowest accessibility. Looking back to the flood simulation



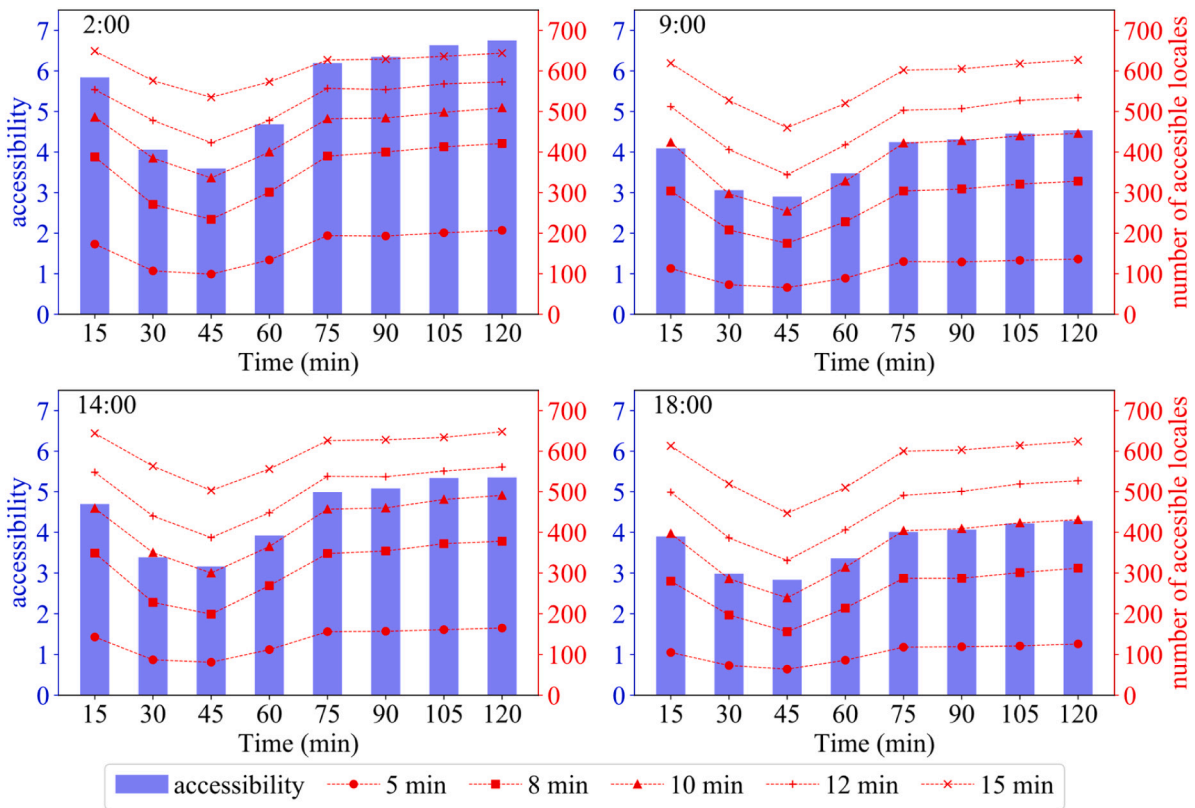


Fig. 6. Variations of EMS accessibility and the number of accessible locations within different timeframes at each 15-minute stage of the flooding.

Table 1

The root mean squared error (RSME) between accessibility in the baseline scenario and other scenarios and the coefficient of variation (CV) of accessibility in all scenarios.

| Scenarios       | Root mean squared error (RSME) |               |               |               | Coefficient of variation (CV) |               |               |               |
|-----------------|--------------------------------|---------------|---------------|---------------|-------------------------------|---------------|---------------|---------------|
|                 | 2:00                           | 9:00          | 14:00         | 18:00         | 2:00                          | 9:00          | 14:00         | 18:00         |
| Normal scenario | <i>Baseline</i>                | 4.6897        | 3.3204        | 5.4161        | 0.7900                        | 0.7652        | 0.7444        | 0.7721        |
| fs - 15 min     | 4.6245                         | 7.2888        | 6.4099        | 7.6528        | 0.8185                        | 0.8441        | 0.8088        | 0.8702        |
| fs - 30 min     | 7.4186                         | <b>8.6601</b> | 8.3146        | <b>8.7076</b> | 0.8747                        | 0.9602        | 0.9131        | 0.9876        |
| fs - 45 min     | <b>8.1183</b>                  | 8.4883        | <b>8.3219</b> | 8.3860        | <b>0.9553</b>                 | <b>1.0413</b> | <b>0.9852</b> | <b>1.0618</b> |
| fs - 60 min     | 6.8453                         | 8.1198        | 7.6779        | 8.0631        | 0.8807                        | 0.9430        | 0.9027        | 0.9610        |
| fs - 75 min     | 5.0572                         | 7.2175        | 6.4164        | 7.5366        | 0.8257                        | 0.8576        | 0.8308        | 0.8859        |
| fs - 90 min     | 4.8180                         | 7.1066        | 6.2836        | 7.4926        | 0.8212                        | 0.8471        | 0.8207        | 0.8731        |
| fs - 105 min    | 4.3994                         | 6.9956        | 6.0408        | 7.3525        | 0.8073                        | 0.8309        | 0.7982        | 0.8540        |
| fs - 120 min    | 4.0932                         | 6.8855        | 5.9000        | 7.2747        | 0.8048                        | 0.8312        | 0.7975        | 0.8546        |

results, we know that the fourth stage has the largest number of road closures, which means the lowest road connectivity during the flooding. But note that to consider the influence of rainfall intensity on travel speed, the parameter  $c$  in Eq. (3) lowers the speed by a larger extent at the third stage than that at the fourth stage, so the accessibility level falls to the lowest point slightly earlier than the time when the lowest road connectivity emerges. To summarize, besides worse road connectivity, lower speed is also an important factor that contributes to the loss of accessibility.

## 5. Discussion on the factors that influence accessibility

This section examines whether there are statistically significant associations between EMS accessibility and the factors we are concerned about. In view of the significant spatial heterogeneity in the accessibility patterns (the values of Moran's  $I$  range from 0.5411 to 0.7972), geographically weighted regression (GWR) analysis is used. Altogether, six latent factors are selected as independent variables, including 1)

population, 2) facility richness, 3) fastest-nearest ratio, 4) real-time speed, 5) road density, and 6) flood risk.

### 5.1. Definition of independent variables

Variable 1 is the population size at a demand location  $i$ . Variable 2 means the number of facilities accessible for location  $i$  within 15 min. Variable 3 means the ratio of the time-shortest path to the distance-shortest path between location  $i$  and facility  $j$  (calculated as  $(L_t/L_{min})/(T_{min}/T_d)$ , where  $L_t$  and  $T_{min}$  are the length and time cost of the time-shortest path,  $L_{min}$  and  $T_d$  are those of the distance-shortest path). Variable 4 and 5 respectively represent the average speed and road density in the minimum rectangle extent determined by the minimum and maximum x- and y-coordinate of the time-shortest path. Variable 6 denotes the time span and length of road closures induced by flood in the same minimum rectangle (calculated as  $\sum(l_k \cdot \sum t)$ , where  $t$  is the flooding time span on road link  $k$ , which takes the value of 1 if link  $k$  is impassable and 0 otherwise).

We build eight GWR models which respectively take EMS accessibility under the normal scenarios (NS-2, 9, 14, 18 in Fig. 5a) and the flood scenarios (FS-2, 9, 14, 18 in Fig. 5a) as the dependent variables. Independent variables for Model 1–4 (normal scenarios) include population, facility richness, fastest-nearest ratio, real-time speed and road density, while independent variables for Model 5–8 (flood scenarios) also include flood risk besides the above five variables. It should be noted that the raw data of all variables are not in the same order of magnitude. For example, the values of the population are much larger than others. Thus, a min-max normalization is applied to the raw data to make them consistently range from 0 to 1.

## 5.2. Results of the geographically weighted regression models

All GWR analyses begin with the Ordinary Least Squares (OLS) regression. The same explanatory variables in a properly specified OLS model are used to run the corresponding GWR model. Moreover, a multicollinearity analysis is performed for each OLS model. Since none of the Pearson coefficients (range from  $-0.4035$  to  $0.6472$ ) for all pairs of independent variables is greater than  $0.7$ , the aforementioned 6 variables are all used. The GWR 4.0 software package (Nakaya et al., 2009) is used to implement the GWR models.

Estimation results for the GWR models are shown in Table 2, including the descriptive statistics (Mean, STD, Min, Max) of independent variables' coefficients, AICc values that indicate the model effectiveness,  $R^2$  and adjusted  $R^2$  that indicate how much variance in EMS accessibility can be explained by the associated factors. Coefficient information about independent variables that are not significant in the OLS model is not displayed in the table. As shown in the table, the adjusted  $R^2$  values are 85.62%, 69.19%, 76.93% and 63.92% for four models in the normal scenarios (Model 1–4), and respectively correspond to a decrease of 13.60%, 16.34%, 17.51%, 15.00% in the variance explained for EMS accessibility at 2:00, 9:00, 14:00, and 18:00 in the flood scenarios (Model 5–8). Besides, each GWR model has a lower AICc value than its corresponding OLS model, indicating a higher level of effectiveness of the GWR models.

Another revealing result is that in the eight scenarios with varying time and flood conditions, significantly associated variables are different. Variable 1 (population) is only significant at 2:00 in the normal scenarios, and 14:00 and 18:00 in the flood scenarios. Variable 2 (facility richness) and variable 5 (road density) are significant in all scenarios. Variable 3 (fastest-nearest ratio) is significant in all normal scenarios, but not significant in all flood scenarios. Variable 4 (real-time speed) is only significant at 2:00 and 14:00 in the normal scenarios. Variable 6 (flood risk) is significant in all flood scenarios. Surprisingly, population and real-time speed are the variables of great concern to us, but they exhibit a trivial association with EMS accessibility.

GWR models address spatial non-stationarity issues, and the coefficients of the independent variables vary in space. Model 1 and Model 5 are selected to display the spatial distribution of coefficients (Fig. 7), since they are the best (highest adjusted  $R^2$  and largest reduction of AICc) models respectively in the normal and flood scenarios. From Fig. 7, we can see that, in the normal scenario (Model 1), population (V1) is positively associated with accessibility for most locations. This outcome is different from a previous study (Xia et al., 2019) which found that, in Tokyo, the EMS accessibility in suburban areas with less population is better than that in central areas with more population. A larger population size means greater competition for service and generally leads to lower accessibility due to the scarcity of service providers. But in this study, the majority of EMS stations are located in areas that are most populated, which to some extent alleviates the service competition pressure. In fact, facility richness (V2) has a positive influence on EMS accessibility, regardless of whether there is flooding or not. Thus, it is important that decision-makers set a budget for new EMS stations. From 2016 to 2020, the Shanghai Municipal Government has demonstrated a commitment to improving its pre-hospital EMS system by building 53

new stations.

An increase in road density can also improve accessibility in most locations except for the northern peripheral areas where roads are sparse. Higher real-time speed (V4) implies that the victims can be accessed by ambulances in a shorter time and naturally result in better accessibility. However, the positive effect of V4 is found only in the western-central part of the study area (Fig. 7 Model 1-V4). In most of the periphery areas, speed is not significant or even has a negative effect on accessibility. Fastest-nearest ratio (V3), which is an auxiliary variable that indicates vehicle speed, is also insignificant or negatively associated with accessibility. This result is different from what we presumed from the beginning and the findings reported in the study of Chen et al. (2016) conducted in Guangzhou. As described in Section 3.3, the highest average speed in the CUA of Shanghai is only 33.17 km/h, indicating that probably the roads are often congested or over-congested. The most significant variations in speed are on the elevated roads or expressways which, however, are less likely to be used by ambulances, since the mandatory response time is only 15 min and ambulances seldom take very long journeys. Therefore, in light of these findings, we suggest that increasing road speed is less effective for improving accessibility in a highly urbanized area with busy traffic.

In the flood scenario, just as anticipated and similar to the conclusions obtained in London by Coles et al. (2017) and Yu et al. (2020), as well as in Shanghai by Yin et al. (2018) and Yang et al. (2020), flood risk (V6) is negatively associated with accessibility. When an extreme flood occurs, more impassable roads or longer road closures can lead to a significant decrease in accessibility. However, both fastest-nearest ratio (V3) and real-time speed (V4) become insignificant determinants for EMS accessibility. Since a considerable number of locations became less accessible or totally inaccessible within the response time due to road closures in the original best routes (time-shortest and distance-shortest), the fastest-nearest ratio and its relationship with accessibility can change drastically. Moreover, because speed is jointly determined by road locations and types (e.g., expressways, primary/secondary/local roads), averaging the speed in the minimum extent of the time-shortest path between a demand location and an EMS station would inevitably smooth the spatial variance, or worse yet, conceal the effect of an underlying factor that is significant for accessibility. Thus, we need to be careful with the transferability of the conclusion in the last paragraph and suggest that the GWR models should be further optimized in terms of the selection of independent variables to obtain a full picture of the factors that influence EMS accessibility.

## 6. Conclusion

This paper showed that, with a case study in Shanghai, some regular (e.g., dynamic population and traffic) and irregular (e.g., extreme flooding) contextual factors in urban areas can significantly influence the measurement of EMS accessibility. By integrating fine-grained time-varying demand and travel cost in the E2SFCA method, we address the limitations of past studies that assumed static population and traffic and thus ignored the effect of uncertain contexts on EMS accessibility. This methodological improvement is the main contribution and value of the paper. In addition, as the study by Ren et al. (2014) found, ignoring contextual uncertainties like variations in population distribution and traffic can lead to serious error in accessibility evaluations, which means that some areas may have unexpectedly low accessibility during certain emergency situations (e.g., earthquakes or nuclear power plant accidents) that may put a large number of needy people at risk. Thus, this study not only develops a methodology for taking some of these uncertainties into account but also highlights why these uncertainties have significant implications for all emergency services in other cities of the world.

Some major findings are as follows. 1) Under non-flood conditions, central areas that are most populated have higher accessibility than peripheral areas with smaller population sizes. 2) In the flood scenarios,

**Table 2**

Estimation results for the geographically weighted regression models.

| Independent variables' coefficients | Model 1 (2:00, non-flood) |        |         |        | Model 2 (9:00, non-flood) |        |         |        | Model 3 (14:00, non-flood) |        |         |        | Model 4 (18:00, non-flood) |        |         |        |
|-------------------------------------|---------------------------|--------|---------|--------|---------------------------|--------|---------|--------|----------------------------|--------|---------|--------|----------------------------|--------|---------|--------|
|                                     | Mean                      | STD    | Min     | Max    | Mean                      | STD    | Min     | Max    | Mean                       | STD    | Min     | Max    | Mean                       | STD    | Min     | Max    |
| Intercept                           | -0.0091                   | 0.1276 | -0.3800 | 0.6291 | -0.0013                   | 0.1096 | -0.2727 | 0.6256 | -0.0093                    | 0.1479 | -0.3877 | 0.5312 | 0.0180                     | 0.1158 | -0.2607 | 0.5471 |
| Population                          | 0.1638                    | 0.1708 | -0.2220 | 0.6022 | -                         | -      | -       | -      | -                          | -      | -       | -      | -                          | -      | -       | -      |
| Facility richness                   | 0.6082                    | 0.2769 | -0.0621 | 1.5212 | 0.5826                    | 0.2713 | -0.1901 | 1.4094 | 0.6472                     | 0.2941 | -0.4406 | 1.6581 | 0.5864                     | 0.2722 | -0.2870 | 1.3685 |
| Fastest-nearest ratio               | -0.5781                   | 0.7256 | -4.9730 | 0.9742 | -0.4498                   | 0.7400 | -3.5175 | 1.9953 | -0.6511                    | 0.7901 | -3.2123 | 1.6398 | -0.7417                    | 0.8687 | -3.6257 | 2.4372 |
| Real-time speed                     | 0.0552                    | 0.2659 | -0.7581 | 1.0158 | -                         | -      | -       | -      | 0.0539                     | 0.3320 | -0.7905 | 1.1406 | -                          | -      | -       | -      |
| Road density                        | 0.7156                    | 0.8830 | -2.1772 | 3.9306 | 1.4034                    | 1.2823 | -1.8675 | 6.2103 | 1.3330                     | 1.1675 | -1.7414 | 6.0508 | 1.6592                     | 1.5788 | -2.8169 | 7.5026 |
| AICc <sup>a</sup>                   | -1398.2 (-205.4)          |        |         |        | -995.1 (-104.5)           |        |         |        | -1007.1 (-162.8)           |        |         |        | -723.2 (-103.5)            |        |         |        |
| R <sup>2</sup>                      | 0.8983                    |        |         |        | 0.7562                    |        |         |        | 0.8293                     |        |         |        | 0.7143                     |        |         |        |
| Adjusted R <sup>2</sup>             | 0.8562                    |        |         |        | 0.6919                    |        |         |        | 0.7693                     |        |         |        | 0.6392                     |        |         |        |

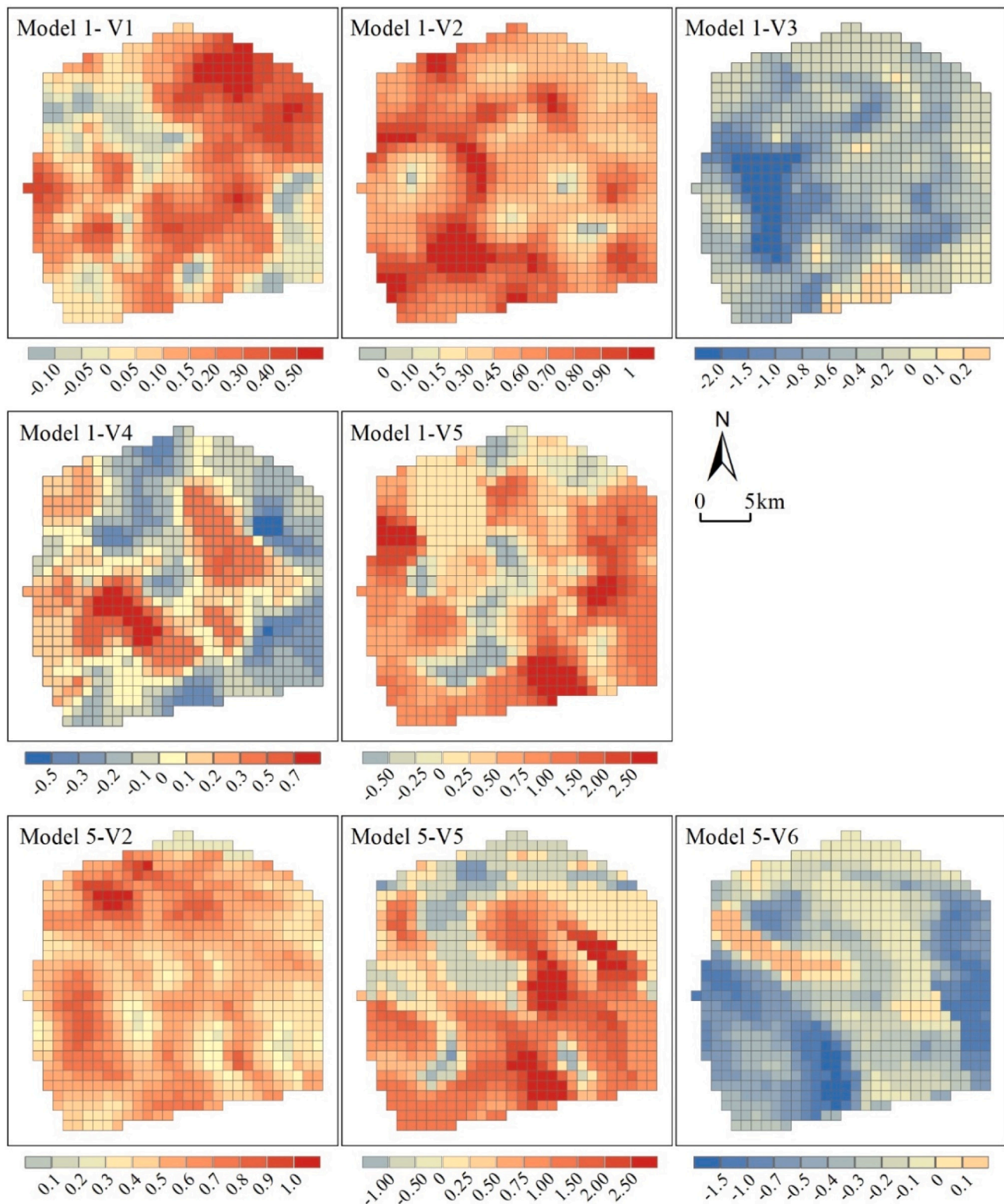
  

| Independent variables' coefficients | Model 5 (2:00, flood) |        |         |        | Model 6 (9:00, flood) |        |         |        | Model 7 (14:00, flood) |        |         |        | Model 8 (18:00, flood) |        |         |         |
|-------------------------------------|-----------------------|--------|---------|--------|-----------------------|--------|---------|--------|------------------------|--------|---------|--------|------------------------|--------|---------|---------|
|                                     | Mean                  | STD    | Min     | Max    | Mean                  | STD    | Min     | Max    | Mean                   | STD    | Min     | Max    | Mean                   | STD    | Min     | Max     |
| Intercept                           | 0.0324                | 0.0874 | -0.2794 | 0.3120 | 0.0345                | 0.0613 | -0.1015 | 0.1972 | 0.0398                 | 0.0685 | -0.1055 | 0.2551 | 0.0415                 | 0.0478 | -0.0581 | 0.1411  |
| Population                          | -                     | -      | -       | -      | -                     | -      | -       | -      | 0.1188                 | 0.3377 | -0.5831 | 1.6645 | -0.0450                | 0.1892 | -0.4538 | 0.5336  |
| Facility richness                   | 0.4902                | 0.1929 | 0.0332  | 1.1055 | 0.4314                | 0.1935 | -0.0102 | 0.8642 | 0.5010                 | 0.1443 | 0.1986  | 0.9088 | 0.5429                 | 0.1449 | 0.3601  | 0.8680  |
| Fastest-nearest ratio               | -                     | -      | -       | -      | -                     | -      | -       | -      | -                      | -      | -       | -      | -                      | -      | -       | -       |
| Real-time speed                     | -                     | -      | -       | -      | -                     | -      | -       | -      | -                      | -      | -       | -      | -                      | -      | -       | -       |
| Road density                        | 0.7646                | 0.9890 | -1.8994 | 4.5069 | 0.9629                | 1.0816 | -1.1697 | 4.1101 | 0.9070                 | 1.0545 | -1.1741 | 4.3210 | 0.7525                 | 0.8054 | -0.7792 | 2.3005  |
| Flood risk                          | -0.3993               | 0.3989 | -1.7884 | 0.5830 | -0.3784               | 0.3146 | -1.4549 | 0.2986 | -0.4081                | 0.3554 | -1.4917 | 0.5314 | -0.3410                | 0.2237 | -0.9206 | -0.0497 |
| AICc                                | -1288.9 (-110)        |        |         |        | -1015.4 (-84.7)       |        |         |        | -980.0 (-79.6)         |        |         |        | -997.1 (-66.3)         |        |         |         |
| R <sup>2</sup>                      | 0.7742                |        |         |        | 0.5811                |        |         |        | 0.6487                 |        |         |        | 0.5251                 |        |         |         |
| Adjusted R <sup>2</sup>             | 0.7203                |        |         |        | 0.5285                |        |         |        | 0.5941                 |        |         |        | 0.4893                 |        |         |         |

- Not significant.

<sup>a</sup> Numbers in the brackets represent the reduction in the value of AICc comparing with the corresponding OLS model.





**Fig. 7.** Spatial distribution of independent variables' coefficients in Model 1 and Model 5.  
(V1: population, V2: facility richness, V3: fastest-nearest ratio, V4: real-time speed, V5: road density, V6: flood risk).

EMS accessibility declines remarkably, and most locations with higher accessibility in the normal scenario may also be more accessible in the flood scenario. 3) On a daily scale, the night time exhibits the best accessibility, followed by the afternoon leisure hours, the morning peak hours, and the evening peak hours successively. On the event scale, the global accessibility level falls to the lowest point slightly earlier than the

time when the worst road connectivity emerges. 4) Statistically, accessibility is affected by contextual factors among the population, facility richness, vehicle speed, road density, and flood risk, but the effects of associated factors are different in the normal and flood scenarios and also vary in space and time of day. Specifically, an increase in facility richness and road density and a decrease in flood-induced road closures

have a significant and positive effect on EMS accessibility.

These findings have important planning implications. The accessibility maps (Fig. 5a) enable decision-makers to identify locations that are less accessible or inaccessible within the mandatory response time and then to take interventions such as locating a new EMS station or relocating existed ones. But this needs to be further studied. We suggest that decision-makers set a budget for new EMS stations. Besides, identifying the worst scenario (e.g., an extreme flood event happens during the evening peak) in which accessibility deteriorates to the largest extent can help them to develop more robust contingency plans, so as to timely rescue victims even in the context of bad traffic or rare accidents. Other modes of transport such as hovercrafts or helicopters are also recommended for areas totally isolated from the surrounding EMS stations in the extreme flood scenario. Moreover, new concepts of urban planning — e.g., low impact development (Pour et al., 2020) — should be encouraged to build a more sustainable and resilient urban environment that can protect its road facilities from flooding when an extreme rainstorm occurs.

To a certain extent, the above recommendations are universally applicable regardless of where the case study is conducted. For example, there is a general consensus about the need for planning new EMS stations in areas of under-provision. However, it cannot be implemented unless we know the details about where the under-served populations are (indicated by locations of low accessibility) and how the shortage of service comes about (indicated by factors that influence accessibility). Sometimes good policies are expected to be very individualized, with appropriate consideration given to the real needs. Thus, instead of promoting specific policies, we are emphasizing the necessity of considering the uncertain contextual effects when measuring EMS accessibility. For the benefit of international practice, decision-makers in other cities need to formulate policies in accordance with specific local conditions to achieve higher EMS performance. Also, since the open access of spatio-temporal big data (e.g., mobile signaling data, real-time traffic data) is increasing, our method can be readily deployed in other scenarios and regions of the world that are concerned about the effects of such uncertain contexts on accessibility.

This study also has some limitations. First, the upper limit of service capability is not considered. All EMS stations are assumed to have adequate professionals and ambulances that can be dispatched to victims as needed. But this may not be the case in reality and may lead to an overestimation of accessibility especially in areas with a denser population. Second, the disparity of morbidity is ignored. A more accurate demand estimation should take into account the morbidity of different kinds of disease at different locations, or at least the percentage of vulnerable groups like the old and the disabled (Yu et al., 2020). These two limitations are also the assumptions based on which this study is conducted. Besides, our population data and traffic data are not collected during the same time period. Although this limitation does not influence the methodology and major findings of the paper, we suggest that, while making policies in practice, one should use the latest data and keep all data consistent in time as far as possible. Lastly, although we considered the effects of rainfall (besides flooding) on vehicle speed, the actual interaction between rainfall and an ambulance's behavior might be much more complex than we can describe. This is a challenging problem to address unless we can crawl very representative traffic data based on which the vehicle speed on rainy days can be well-calibrated. The accessibility results also should be further validated if we can have access to real EMS calls data on days with extreme rainfall and flooding.

#### CRediT authorship contribution statement

Mengya Li designed the initial method, performed the data processing and analysis, interpreted the results and wrote the first draft of the manuscript. Mei-Po Kwan performed the reviewing and editing and contributed to the further development of the methods and interpretations. Mengya Li and Mei-Po Kwan together performed the

revision according to the comments provided by the editor and the reviewers. Jie Chen performed the estimation of population based on mobile signaling data. Jie Yin and Dapeng Yu contributed to the simulation of the flood scenarios. Jun Wang contributed to the conceptualization and coordination of this work.

#### Declaration of competing interest

The authors declare no competing financial interests.

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